

1] What is machine learning?

Machine learning (ML) is a branch of Artificial Intelligence that enables system to learn from data (to predict future values) & improve performance without explicit programming.

2] What are the different types of machine learning?

- i) supervised learning - learns from labeled data
- ii) Unsupervised learning - finds patterns in unlabeled data
- iii) Reinforcement learning - learns via rewards & penalties.

Supervised $\begin{cases} \text{classification} \\ \text{Regression} \end{cases}$

Unsupervised $\begin{cases} \text{clustering} \\ \text{Association} \end{cases}$

Semi-supervised learning : learns from mix of labeled + unlabeled data

3] What is difference between AI & ML?

Feature	AI	ML
Def	Broad field of intelligent systems	subset of AI
Goal	Mimic human intelligence	Learn from data
Scope	Wide (robots, NLP, vision)	Focused (data-driven models)
Ex	Chatbots	Spam email classifier

4] Lifecycle of ML :-

- i) Problem definition
- ii) Data collection
- iii) Data cleaning & Data preprocessing
- iv) EDA
- v) Feature engineering
- vi) Model selection
- vii) Model training
- viii) Model evaluation
- ix) Model deployment
- x) Maintaining & monitoring

5] What is structured data?

The data organized in a fixed format.

structured data is data arranged in rows & columns (like a table).

Ex: Excel, SQL database

6] What is unstructured data?

It has no fixed format.

Unstructured data is raw data that has no predefined schema.

Ex: Images, videos, text

7] What is supervised learning?

It is a type of ML that uses labeled data to train models.

Ex: Spam Email detection

8] What is unsupervised learning?

It is a type of ML that finds patterns in unlabeled data.

Ex: Grouping customers based on buying behavior.

9] What is model selection?

Model selection is choosing the best algorithm for a given problem.

10] Training Data:

It is the dataset used to teach the model.

11] Testing Data:

It is used to evaluate model performance on unseen data.

12] Confusion matrix:

It is a table used to evaluate classification performance.

	Actual positive	Actual negative
Predicted positive	True Positive	False Positive
Predicted negative	False Negative	True negative

13] Correlation :

It measures the relationship between two variables

Range: -1 to +1

Ex: Temp v/s Ice cream sales

14] Why regularization techniques are important?

It reduces overfitting by penalizing complex models

- Prevents overfitting
- Improves model generalization

Types :- L1 (Lasso)

L2 (Ridge)

15] How to handle missing values?

It is a process of treating missing data in a dataset

Ex: Filling missing values with mean / median / mode

16] How to detect the outliers?

i) Box plot Method (IQR method):

calculate: Q_1, Q_3

$$IQR = Q_3 - Q_1$$

Formula: lower bound = $Q_1 - 1.5 \times IQR$

Upper bound = $Q_3 + 1.5 \times IQR$

Values outside these range = outliers

ii) Z-score Method:

$$Z = \frac{x - \text{Mean}}{\text{std deviation}}$$

If $|z| > 3 \rightarrow$ outlier

iii) Scatter Plot Method

iv) Using ML (Isolation Forest, DBSCAN, One-class SVM)

17] What are the advantages of preprocessing?

- Improves accuracy
- Removes noise
- Handles missing data
- Normalizes values

18] What are the different data sources available to collect the data?

- (i) Internal data sources (within org): Databases (MySQL, Oracle), CRM systems, ERP systems, Sales records, Employee data.
- (ii) External data sources (~~outside~~ outside org): APIs, Government datasets.

Other sources - surveys, kaggle, social media etc.

19] What is feature selection?

Selecting important features to improve performance

20] How to maintain & monitor the data?

- Data versioning
- Regular retraining
- Monitor model drift
- Logging & alerts

21] What are different techniques available in supervised learning?

- Linear regression
- Logistic regression
- Decision Trees
- Random forest
- Support Vector machine (SVM)
- K-Nearest Neighbors (KNN)

22] Applications of ML:

- (i) E-commerce recommendation systems: product suggestion
- (ii) Healthcare: Drug discovery, disease prediction
- (iii) Banking & Finance: fraud detection, credit score prediction, stock market prediction
- (iv) Self-driving cars: Object detection
- (v) Social media: face recognition, sentiment analysis
- (vi) Entertainment (OTT platforms): movie/series suggestion
- (vii) Weather forecasting
- (viii) Manufacturing
- (ix) Business Insights
- (x) NLP appl'n

23] History of ML:

- * 1940s-1950s: Alan Turing introduced idea of intelligent machines. proposed the Turing test.
- * 1959: Arthur Samuel coined the term machine learning. Developed a self-learning checkers problem
- * 1960s: Development of Perceptron (early neural network)
- * 1970s - 1980s: Rise of Expert Systems
Followed by AI winter due to lack of progress
- * 1990s: Shift to data-driven approaches. Algo like SVM, DT
- * 2000s - Present: Growth of Big Data & DL

24] Evolution of ML:

- i] Rule-Based Systems: Machines followed hardcoded rules.
No learning capability
- ii] Traditional ML: S/m learn from structured data
Algo - Decision Trees, Naive Bayes, SVM
- iii] Statistical learning: focus on probability & mathematical models
- iv] Big Data Era: Availability of large datasets
Faster computation using cloud & GPUs
- v] Deep learning & AI: Use of neural networks with multiple layers.